

Unlocking the ZYXEL AX7501-B0: Overcoming ISP Firmware LOCKS

Target: Zyxel AX7501-B0 (BCM6858, ARM32 LE), ISP-locked `V5.17(ABWW.5)C0`.

Goal: flash stock Zyxel `V5.17(ABPC.8)C0` (or any later ABPC).

Result: unlocked router with ZYXEL firmware and CFE upgrade path.

The Zyxel AX7501-B0 is a high-end GPON/XGS-PON router designed for fiber connections, capable of 10GB throughput with an SFP+ port. ISPs often buy these in bulk and flash their own locked firmware onto them before giving them to customers - disabling SSH, firmware upgrades, and any real admin access so you can't mess with their configuration. I bought one off the marketplace and it came with one of these ISP locked firmwares on it. This is how I unlocked it.

if your ABWW-locked AX7501-B0 has no upload page and no SSH, this is for you.
if SSH or the upload page works skip to Lock 1.5.

Lock 1 uploading firmware on the router :

Step 1.1 actually communicating with the Zyxel Router

The first lock I encountered was communicating with the router to upload firmware. Some people might have a version of the WebGUI that allows them to upload the firmware there I did not. (located on the WebGUI in Maintenance -> Firmware Upgrade)

The next option would have been to connect the router to my computer directly through an ethernet cable and ssh to it. Some routers may allow this, mine did not you can test if your router allows this by connecting the router to your computer with an ethernet cable and launching the following command in a cmd/powershell window (SSH admin@192.168.1.1) if you manage to connect please jump to (Step 1.5)

My unit (ABWW.5) had the web upgrade page removed and all local management ports were disabled and it was not possible to reactivate them through the web gui since the page that would possibly have allowed that was removed as well.

The only remaining entry point was hardware (i was thinking dumping the nand chip patch the firmware and reuploading it)

After some research online i found out about UART pins that were located inside the router
To access them proceed to unscrewing the 4 screws found on the bottom of the router and carefully popping the plastic latches off.

You will then be able to see the motherboard (look at the following page) image 1

If you look around the middle of the motherboard you will find 4 pins (look at image 2)

Your pins should look like this (O represents a pin X represents the empty pin slot) OOOXO

I confirmed the research made by others online with a multimeter by checking the pin that was alone to confirm it was the ground pin.

(you can check image 2 if you need visual reference)

I personally opted for using a RaspBerry PI 3 Model B V1.2 since i do not own a UART to USB connector) Then using Dupont connectors i connected the Ground Tx pin(gpio 14) and Rx pin(gpio 15) to the router (check image 3 for visual reference)

(Do not forget to connect Rx->Tx and Tx->Rx from Pi->Router)

Next Step trying to get into CFE shell to flash

Step 1.2 Trying to get into CFE shell

After properly wiring the PI to the Router and testing that the pins did connect properly with my multimeter i ssh'ed into the Pi and executed the following commands

`tmux new -s SL` (to be able to reconnect and see logs in case my pi dropped connection or for any other reason)

`sudo picocom -b 115200 -d 8 -p n -f n -g /log.txt /dev/ttyAMA0` (to be able to log all the debug in /log.txt and control the router through the debug pins)

After that i plugged the router in and pressed the power button

And got spammed with a LOT of debug prints

After reaching this line however ("Hit any key to stop autoboot:005") i pressed a key to get into the CFE shell with hope i could flash the firmware on there sadly it prompted me for a password

I did not manage to find which one it was even after testing multiple "default" ones

(1234/admin/supervisor/0000/The default WPA key / admin password printed on the bottom sticker of the router. None of these were accepted.)

It might be one of those for you if it is skip to step ("User Error 1.2 CFE shell usage")

Step 1.3 Finding another solution

I was getting kind of desperate and frustrated at this point. But i kept digging

I thought to myself i would let it boot completely and read the debug logs maybe they would reveal something interesting

After letting it completely boot i pressed enter by accident and the router prompted me for an username ("AX7501-B0 Username:") i entered admin and then it prompted me with a password ("Password: ") i entered the same password i used to connect to the webgui ("192.168.1.1") and i worked i got into some sort of shell ("ZySH>") it does seem very limited but still gives me everything i needed to start testing

Step 1.4 Activating SSH/TELNET/FTP

I started digging around to see how i could upload firmware / what i could unlock and found out i could run those 3 commands

```
zycli mgmtsrvctl config --service SSH 0
zycli mgmtsrvctl config --service FTP 0
zycli mgmtsrvctl config --service TELNET 0
```

Those 3 activated SSH FTP and TELNET

Step 1.5 Uploading the firmware and the start of the problems

after downloading the firmware (i found the one i used here : <https://www.studerus.ch/de/support/download/60745>) i proceeded to try to upload the firmware to the router by doing the following : Connecting the router directly to my pc with an ethernet cable and doing the following commands in a command prompt shell : ("ftp 192.168.1.1"), ("bin"), ("cd fw"), ("put C:\Path\to\firmware.bin fw.bin")

I then saw the power light of the router flashing red and though it was flashing the firmware I thought i was done well how wrong i was after leaving my room doing some other work and coming back 5-10 minutes later it was still flashing red

This might not happen to you, maybe it will flash without any issues and you will have a clean installation of the firmware. If that is your case, lucky you if it is not i will let you continue reading.

Lock 2 disabling the first checks

(source : ("Zyxel | Hack GPON <https://hack-gpon.org/zyxel/> "))

Disable firmware version check and model check by running the following commands.

```
zycli fwidcheck off
zycli modelcheck off
```

Lock 3 128KB block size firmware

I was not sure what was going wrong so i went back to my raspberry pi tmux session and found the following error

("ERROR: NAND flash block size 256KB does not work with an image built with 128KB block size")

I proceeded to ask on forums and the zyxel support directly if they could provide me with a 256KB block size version of the firmware and in the meantime i continued looking for solutions on my side

I continued digging around in the ZySH shell hoping there was something that could help me (there was not or at least i didn't find it)

After a bit of looking online tho i learned that i could unpack and repack the firmware myself with 256KB blocks since it was an UBI image

Using ("~/local/bin/ubireader_extract_images -o extracted V517ABPC8D0.bin ") on my WSL instance i unpacked the raw .ubifs files

And using ubinize ("ubinize -o V517ABPC8D0_256k.bin -m 2048 -p 262144 ubi.cfg") i was able to reconstruct a 256kb version of the firmware which i hoped would be enough to finally flash a working version of the firmware to my router...

Lock 4 Illegal ZyXEL image header of whole flash image!

I flashed the firmware again using ("ftp 192.168.1.1"), ("bin"), ("cd fw"), ("put C:\Path\to\firmware.bin fw.bin") and let it flash for 3-4 minutes after i saw the power light was still flashing red i went to my tmux session again to see if a different issue appeared.

Yes there was a new error

Illegal ZyXEL image header of whole flash image!

Firmware Upgrade fail

Zyxel's firmware seems like it has a proprietary header/metadata somewhere that was lost in the repacking

I compared both the repacked version and the original version

wfi_tag offset field value

+0x00 wfiCrc 0x3f008aea

+0x04 wfiVersion 0x00005732

+0x08 wfiChipId 0x00006858

+0x0C wfiFlashType 0x00000003

+0x10 wfiFlags 0x00000002

+0x14.. boardId "PX7501-B0"

+0x34.. version "V5.17(ABPC.8)C0"

+0x3FE zyxelChecksum 0x7d76

The last 1024 bytes contained text such as PX7501-B0 and V5.17(ABPC.8)C0

So i just made a python script to take them from the original and place them on the repacked version (i did a lot of other test that are not worth explaining/mentioning since they did not produce results but this one seemed to stick)

Lock 5 Could not determine image format [84672532 bytes]

Before trying anything else i just appended a larger check (instead of 1024 bytes i did 2048)
And that time it did manage to read properly the tag and print it in the debug window

```
zcmdReqFirmwareUpgrade : wfiFirmwareVersion V5.17(ABPC.8)C0
zcmdReqFirmwareUpgrade : wfiKernelChecksum 00001623
zcmdReqFirmwareUpgrade : wfiRootFSChecksum 000015de
zcmdReqFirmwareUpgrade : FirmwareVersion V5.17(ABWW.5)C0
zcmdReqFirmwareUpgrade : KernelChecksum 0000deef
zcmdReqFirmwareUpgrade : RootFSChecksum 0000189e
```

It still threw the error `ERROR[cmslmg_validateImage.912]: Could not determine image format [84673556 bytes]` but we got a little further the `wfi_tag` was being read correctly but `cmslmg_validateImage` was still failing

After some more research i did not manage to find anything further and temporarily gave up trying to find a way to bypass it through firmware and researched if it was possible through hardware with the nand I found what i assume this router uses as its nand ("Macronix MX30LF4G28AD-TI")
I found the following info online

Manufacturer: Macronix
Type: Parallel NAND Flash (SLC)
Capacity: 4 Gbit = 512 MB
Package: TSOP-48
Page size: 2048 bytes + 64 bytes spare
Block size: 128KB

After restarting to look to fix through firmware without touching the hardware i started reversing the code of the firmware (V5.17(ABPC.8)C0) that i was trying to upload hoping it had the same locks as the one on the router and so the solutions

Function: `cmslmg_validateImage` (at 0x10d70)

Behavior: The upgrade routine calculates a CRC32 across the entire image body up to the last 20 bytes (`len - 20`) using Broadcom's `genUtl_getCrc32(buf, len - 20, 0xFFFFFFFF)`

The Quirk: Standard Python/POSIX `zlib.crc32` applies an inverted final XOR (^ 0xFFFFFFFF) Broadcom's implementation is a raw shift-register calculation with no final XOR. To match Broadcom's value in Python, you must invert the result: `zlib.crc32(buf) ^ 0xFFFFFFFF` i verified my `crc32` by running it against the original firmware's stored `crc` before trusting it on the repacked body

The resulting 4-byte CRC is stored in little-endian format in `wfi_tag[0..4]`

Function: `zcmdValidateImage` (at 0x370e8)

Behavior: It computes an additive 16-bit fold checksum (`zyxelChecksum`) across the first 1022 bytes of the tag (`wfi_tag[0..0x3FE]`) and compares it against the stored value at offset 0x3FE

If `zyxelChecksum(tag[0..0x3FE]) != *(u16*)(tag + 0x3FE)`, the device aborts the upgrade

Solution

Extract the intact 1024 bytes wfi_tag trailer from the original firmware

Calculate the Broadcom CRC32 of the repacked 256KB UBI Image body and write it to tag[0..4]

Recompute the 16-bit Zyxel checksum across tag[0..0x3FE] and write it to tag[0x3FE..0x400]

Append the patched trailer to the repacked body

Lock 6 flash block size 256KB does not work with an image built with 128KB block size (again)

After passing all the precedent checks to finally be able to flash (or so i thought) it gave me this error again

I thought i didn't upload it properly or didn't repack it properly but after retrying a few times i did

So i kept looking

Maybe a flag somewhere tells the router what size it is

After confirming ("PEB size: 262144 bytes") was really 256KB i started looking again

When reversing bcmNandImageSet it did not read UBI spacing

I was reading ...

wfi_tag (yes again)

```
#define WFI_NOR_FLASH          1
#define WFI_NAND16_EBI_FLASH  2
#define WFI_NAND128_EBI_FLASH 3
#define WFI_NAND256_EBI_FLASH 4
#define WFI_NAND512_EBI_FLASH 5
#define WFI_NAND1024_EBI_FLASH 6
#define WFI_NAND2048_EBI_FLASH 7
```

Wifi flash type being at 3 i replaced it to 0x00000004

And? IT FLASHED

I will bless you with the flash logs

File Create...

File Upload
Not a json format file
recv_str=/var/fw/fw.bin
Image path /var/fw/fw.bin
Image length 85722112
nvramData->FeatureBits 4 5 7 1
wfiTag->FeatureBits 4 5 7 1
Wi-Fi chip info: 0x43684
CheckingList=0x6715-[ABWW.4]b1
zcmdReqFirmwareUpgrade : wfiFirmwareVersion V5.17(ABPC.8)C0
zcmdReqFirmwareUpgrade : wfiKernelChecksum 00001623
zcmdReqFirmwareUpgrade : wfiRootFSChecksum 000015de
zcmdReqFirmwareUpgrade : FirmwareVersion V5.17(ABWW.5)C0
zcmdReqFirmwareUpgrade : KernelChecksum 0000deef
zcmdReqFirmwareUpgrade : RootFSChecksum 0000189e
kill OneConnect processes to release CPU
killall: one_connect_monitor: no process killed
killall: l2command: no process killed
killall: plctool: no process killed
killall: amptool: no process killed
Stop voice service.
Stopping ZylMS VoIP...
[Dbg] TIME_DELAY_STOP = 0
[Dbg] RCM_DSP_AND_PHONE_ENDPOINT_KEY_PROCESS_CNT = 6
[Dbg] RCM_DSP_AND_PHONE_ENDPOINT_KEY_PROCESS_RUNNING_CNT = 0
[Dbg] IS_TO_DSP_AND_PHONE_DEINIT_DELAY_ONLY_WHEN_ALL_KEY_PROCESSES_RUNNING
= 1
[Dbg] IS_TO_DSP_AND_PHONE_DEINIT_DELAY = 0
[Dbg] IS_EVER_DSP_AND_PHONE_DEINIT_DELAY = 0

No [voiceApp] module is running, do nothing for it...
No [MM] module is running, do nothing for it...
No [IPTK] module is running, do nothing for it...

Sending SIP de-register...
sh: can't kill pid 3838: No such process
2024-02-01 01:02:26 fwwatcher<1386>, zcfg_msg.c:245, _msg_rcv(), WARNING: EID = 20, select()
wait forever, EID = 1, type = 7 with reply, oid = 22322
Sending ppp disconnect and dhcp/dhcp6 release...
killall: dhcp6c: no process killed
killall: pppd: no process killed
killall: udhcpd: no process killed
killall: pppd: no process killed
*** To do runtime data partition recover ***
Backup configure file success !!
Start runtime recover

Erasing 256 Kibyte @ 0 -- 0 % complete
Erasing 256 Kibyte @ 300040000 -- 0 % complete

.. (shortened for logical reasons)

A lot of spamming happened after it (each time the led changed state for example (a lot))

User Error

User Error 1.1 wrong page size

After finally flashing a non ISP locked image to my router i went back to my tmux session and looked at the logs and saw it seemed like it was still booting so i left it run for a bit then realised it was the third time i saw the same print again i then proceeded to shut down the router after realising it was bootlooping over and over

Plugging it out for 30 seconds and back in

And going back to my tmux session just to realise it was still bootlooping

At the moment this happened i thought i bricked the device since i could not get into the limited ("ZySH") shell nor could i ssh/ftp into the router since it never really fully initialized

As a last hope resort i restarted the router and in tmux when i saw "Hit any key to stop autoboot:005" i pressed enter

I didn't have high hopes and thought it would ask me for a password again but reflashing it seems like it removed the CFE password

I then saw the ("CFE>") text and celebrated

User Error 1.2 CFE shell usage

I looked around in the commands and read online about what was doable and landed on the following command ("ATUR") ATUR is a command that allows you to upload firmware and flash it to the router using the Zyxel directly

The utilisation is the following

("ATUR 192.168.1.33:fw.bin")

ATUR is the command

192.168.1.33 is the ip of TFTPd

fw.bin is the name on your computer of the bin file to flash

note: ATUR silently fails past ~88 MiB, see 1.3 if your image is bigger

I decided to use Tftpd64 by Ph. Jounin (source : [TFTPd64 : an opensource IPv6 ready TFTP server/service for windows : TFTP server](#)) look at image 4

I don't exactly know or have done much research on how this works nor on how you are supposed to set it up

But set current directory to the folder where your firmware bin file is found

From what i read on how to use this tool you are supposed to do some setup in your settings i did not completely understand but i will let you do your research since i do not wish to tell you anything wrong

the PC running TFTP64 needs to be manually set to a static IP in the same subnet (e.g., 192.168.1.33 / 255.255.255.0

But basically after that you just place your firmware in the folder you selected in tftpd64 (you need a direct link between your router and your pc through an ethernet cable) then run ("ATUR 192.168.1.33:fw.bin") in your shell and it should upload and flash the firmware you wish to flash

User Error 1.3 File size

After fixing the page size from 2048 to 4096 bytes repacking once more and uploading it to the server with the ATUR command it failed

In TFTP64 i saw that the upload failed at 97% i was unsure why i restarted the router my pc changed the timeout settings tried with python it always failed

Then i saw that the issue was that the max for the ATUR command was a file that was 92,274,688 bytes Anything bigger was too big and could not be stored and flash to the router and the file i built was 95,683,584 bytes
 $95683584 - 92274688 = 3408896$ so it was 3,408,896 bytes too big (3.4MB)

I thought about checking if there was any padding around the end of some sort but didn't find any

mkfs.ubifs sizes the journal at $\text{max_leb_cnt} * \text{LEB} / 8$ by default with -c 900 and LEB=253952 that is a 28 MB journal

So just reducing the -c command from 900 to 400 was enough to reduce the file size enough to around 84MB

After a few more tries of flashing and fixing having to change packing flags
I flashed the image to the router one last time

And finally got into a clean non ISP branded login page when going on ("<http://192.168.1.1/login>")

Last issue...

I immediately tried to connect with admin/admin and admin/sticker password, but this remains an open issue/unresolved.

None of them worked

I am unsure of why

I went back in the shell and could login into the ("ZySH>") shell with the sticker credentials

This shell has most options modifiable through there directly through the config so i did not dig any further to see how to fix that bug because it did not seem like such an issue to me
Personally i will keep it like that since everything works my pc now has 10GB optic fiber i do not care much more to fix this
If you wish to fix this for yourself feel free the router is now unlocked it should not be impossible

SFP+ Information

Originally when plugging in my SFP+ connector the router did not recognise it at all it only recognised the one that came with the router

After flashing it printed a debug print that it did not recognize the SFP+ device plugged in and would switch to default mode instead of staying silent and default mode seems to work fine for me

IMAGE #1

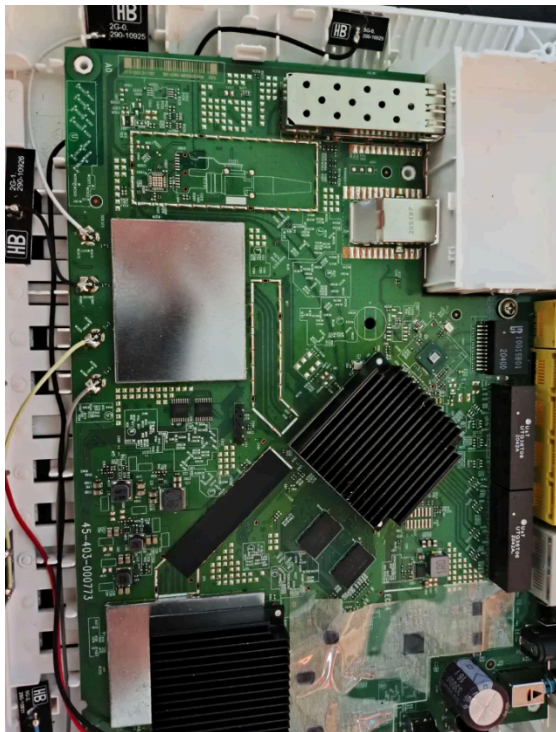


IMAGE #2 (source [Zyxel | Hack GPON](#)) the image i took was very blurry and when writing this i have

Tftpd64 by Ph. Jounin

Current Directory: C:\Program Files\Tftpd64

Server interfaces: 127.0.0.1 Software Loopback Interface 1

Tftp Server | Tftp Client | DHCP server | Syslog server | Log viewer

peer	file	start time	progress	bytes	total	timeo...

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